(Please write your Enrollment Number)

Enrollment No. 0160117 2021

MID-TERM EXAMINATION

Course Name : B. Tech. CSE-AI Semester : 6

(March, 2024) OFF LINE mode

Subject Code: BAI-322	**Subject: Data Warehousing and Business Intelligence**
Time : 1 ½Hours	**Maximum Marks : 30**
Note: Q. 1 is compulsory.	

Q1		**(2.5*4)**	
	(a) Differentiate between a Database System and a Data Warehouse.	UNIT-1	
	(b) Why is there a compelling need for data warehousing in modern business environments?	UNIT-1	
	(c) What is a Fact Constellation schema?	UNIT-2	
	(d) What are some considerations when implementing OLAP in a Data Warehouse?	UNIT-2	
Q2	**(Attempt any Two Parts) UNIT-1**		**(5,5)**
	(a) Analyze the components of a Data Warehouse, providing an overview and emphasizing the importance of the Three-Tier Architecture in its structure.		
	(b) What is dimensional analysis, and how does it contribute to defining business requirements?		
	(C) Discuss the idea of information packages within the framework of Data Warehousing.		
Q3	**(Attempt any Two Parts) UNIT-2**		**(5,5)**
	(a) Compare and contrast the STAR schema and Snowflake schema as different types of schemas in Dimensional Modeling.		
	(b) Explore OLAP operations such as drill-down, roll-up, slice-and-dice, and pivot or rotation. Provide an overview of OLAP models, emphasizing variations, and discuss the MOLAP model versus the ROLAP model.		
	(c) Explain the objectives of Dimensional Modeling and the process of transitioning from requirements to data design in the context of Data Warehousing.		